Day-76

21 Oct, 2024

VLANS (Part I)

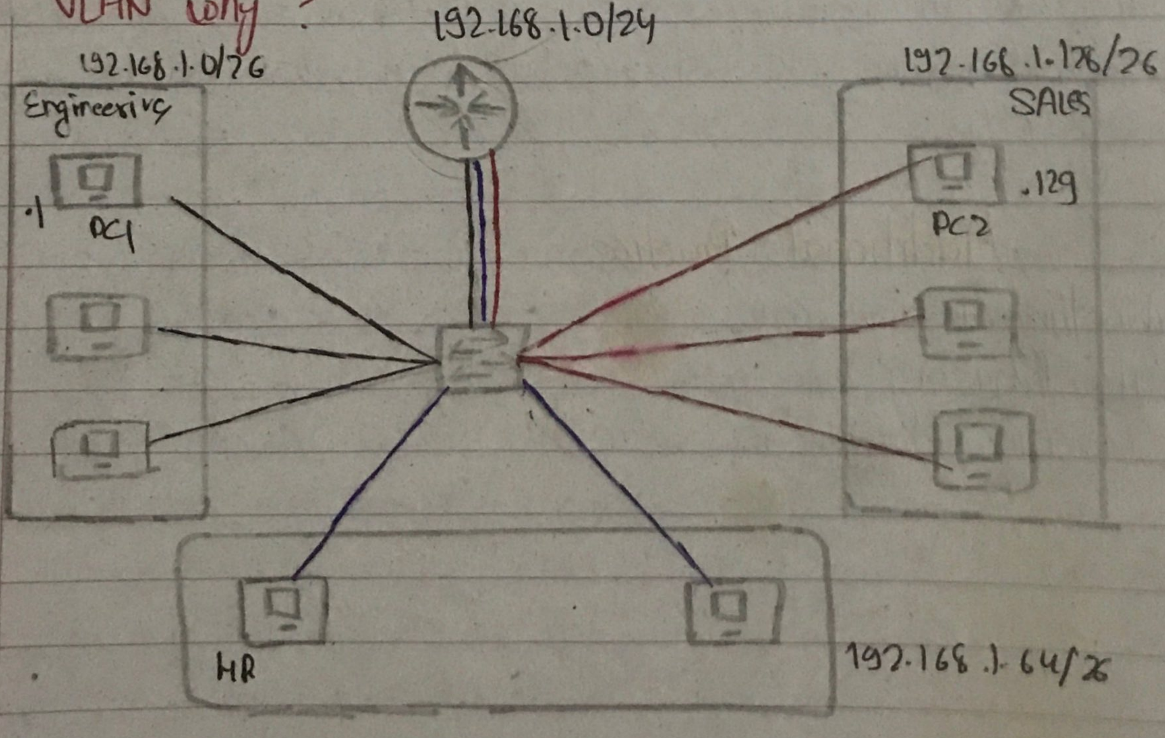
LAN

A LAN is a group of devices (PCs, Servers, routers, switches, etc) in a single location (home, office etc)
A LAN is a single broadcast domain, including all devices on that brodcast domain.

Broadcast Domain

A broadcast domain ~~in~~ is the group of device which will receive a broadcast frame (dstn MAC add - FFFF.FFFF.FFFF] sent by any one of the members.

VLAN why ?


192.168.1.0/24
192.168.1.0/26
Engineering
.1
PC1
192.168.1.128/26
SALES
.129
PC2
HR
192.168.1.64/26

The company is using 192.168.1.0/24 network for this LAN. This ~~a~~ isn't necessarily the best setup for both security & performance purpose. It would be best to split up these into separate subnets. For eg. if PC in Engg. depart sends a broadcast mssg intended for other PCs in Engg. depart. Since its, broadcast, switch will flood it through all interfaces & PCS in other departs as well as Router would receive broadcast. It is bad for security & network performance.

Performance: Lots of unnecessary broadcast traffic can reduce network performance.

Security : Even within same office, we want to limit who has access to what. We can apply security policies on a router/ firewall.

Bcoz, it's one LAN in our case, PCs can reach each other directly, without traffic passing through router. So, even if we configure security policies on Router / firewall it won't have any effect.

Now, when we provide subnet to each depart, then too there's problem. Router is going to need IP add for each subnet, So it will need one interface per subnet

If PC1 sends some data to PC2, PC1 will recognize that PC2 is in different subnet than its own, so it will set the destination MAC to its default gateway R1

PC1 will send packet to R1 through SW, with dstn MAC of R1. R1 receives it and send it to PC2 through SW, with dst mac of PC2 and src mac of R1

So, instead of sending traffic directly to PC2, we forced it to send traffic through Router where we would have configure security policies.

There's a still a problem.

What if frame is broadcast or unknown unicast?
↳ The switch will flood frame out of all interfaces
Switch is only aware upto layer 2, It looks only at
layer 2 ~~interfaces~~ information like src & dst MAC dst.
It doesn't care about layer 3,4 & ~~upper~~ other layers.

This is bad both in terms of security & network performance. Though we separated three departments into three subnets, they are separated at layer 3 but they are still in same broadcast domain. (Layer 2)

One possible solⁿ is to buy separate switch for each department but hella expensive way.

This is where VLAN comes in

VLAN

Although these PCS are at same LAN, we can use VLAN to separate them at layer 2.

Er depart = VLAN10
HR depart = VLAN 20
SALES depart = VLAN 30

We configure VLANS on switch interfaces.
We configure the switch interfaces to be in ~~VLAN~~ a specific VLAN, and the end host connected to that interface is part of that VLAN.

The switch will consider each VLAN as separate LAN & won't forward traffic between VLANS, not even broadcast or unknown unicast.

Router is used to route between VLANS. The

Scanned with CamScanner

Switch doesn't perform inter VLAN routing.

Traffic that ~~comes~~ arrives ~~from~~ on VLAN 10 interface is forwarded out of VLAN 10 interface.

What is VLAN

- VL are configured on switches on a per-interface basis.
- logically separate end hosts a layer 2.

Basic VLAN configuration

i) `show vlan brief` {~~sho~~ similar to show ip routes or show ip int brief}

ii) VLAN 1 is default. All interfaces are assigned to VLAN 1 by default.

VLAN 1002-1005 ~~are~~ are used for FDDI & Token ring.

They are old tech & we don't need it for CCNA.

So, there are five VLAN ~~at~~ beginning by default & can't be deleted

ii) `interface range g1/0-3` {select interfaces ~~to~~ combining which we wanna keep in VLAN}

iii) `switchport mode access`

iv) `switchport ~~no~~ access vlan 10`

(Untagged ports) Acces port: Switchport which belongs to a single VLAN.

(Tagged ports) Trunk ports: Switchports which carry multiple VLAN.

A switchport connected to end host should enter access mode by default. But still its good to configure manually.

If vlan 10 wasn't there, it will get created automatically ~~still~~ when assigning it to interface

v) vlan 10
(similar to interface g0/0.) इथे vlan 10 [illegible] [illegible] Property edit करते मिळह, like name. also if vlan 10 [illegible] vlan [illegible] [illegible] it will get created

QUIZ

Q.3) What happens if you try to assign a switch ~~[illegible]~~ interface to a VLAN that doesn't exist.
a) Command will fail.
b) [✓ switch will create VLAN]
c). Interface will be disabled until you create the VLAN.
? d) ALL VLANS exist by default.

Q.5)- You create VLANS 10, 20 & 30 on a CISCO Switch. How many VLANS will be displayed in the output of the show VLAN brief command.?
a) 3   b) 5   [✓ c) 8]   d) 10.